John Bobby

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.12
Kuriakose Elias English Medium School Mannanam Kottayam Kerala	Class XII (DHSE Kerala)	2024	99.70%
Guardian Public School Mattapilly Ernakulam Kerala	Class X (CBSE)	2022	97.80%

ACHIEVEMENTS

ISRO VLSI Problem Statement | Inter IIT Tech Meet 14.0 (6th Place)

- Worked on a **Delta Sigma ADC** design as part of the ISRO VLSI problem statement.
- Designed and evaluated **Continuous-Time** and **Discrete-Time Modulators** in **MATLAB/Simulink.**
- Evaluated the impact of **oversampling ratio** and **sampling frequency** on **SNR** and **ENOB**.
- Secured **6th place nationally** in the problem statement.

PROJECTS

Dual-Core SIMT GPU with Warp Scheduling and Branch Divergence Handling | **Verilog HDL**

- Designed a **dual-core SIMT GPU** with **4 warp schedulers/core** supporting **16 threads/warp**.
- Implemented a **round-robin warp scheduler** and **queue-based memory scheduler** for efficient execution.
- Implemented **mask-based branch divergence handling** for correct execution of divergent control-flow paths.
- Verified functionality using Verilog simulations and GTKWave.

RISC-V RV32-I Processor Design | **Verilog HDL**

- Designed and implemented a **5-stage pipelined 32-bit RV32I processor** in Verilog.
- Integrated **hazard detection** and **operand forwarding** to resolve pipeline data hazards.
- Implemented a **BHT+BTB branch predictor** for dynamic branch prediction.
- Verified the design using Verilog testbenches and GTKWave.

MNIST Digit Classifier - CNN Accelerator | Verilog HDL

- Designed a **fully streaming CNN accelerator** for **MNIST digit classification** in Verilog HDL.
- Integrated a controller, **4 line buffers, 8 convolution-ReLU, max-pooling,** and **10 fully connected neurons**.
- Implemented fixed-point inference using **PyTorch-trained weights**.
- Completed RTL design for all modules; simulation and testing pending.

FIR Filter and FFT Processor on FPGA | **Quartus Prime Lite**

- Developed **100-tap fixed-point FIR filters** in Verilog with pipelined and non-pipelined architectures.
- Compared **resource utilization, critical path delay, and achievable Fmax** across both FIR implementations.
- Implemented a **pipelined 8-point Radix-2 FFT processor** in Verilog.
- Verified functional correctness using **MATLAB**-based reference simulations.
- Synthesized and deployed both designs on the **Intel DE10-Lite FPGA** using Quartus Prime Lite.

Dual-Channel USB Oscilloscope | **Altium Designer**

- Designed the complete **schematic** for a **dual-channel USB oscilloscope** around an **STM32 microcontroller**.
- Integrated the **ADC interface** and **USB communication** for real-time waveform acquisition.
- Designed the complete **PCB layout** in Altium Designer.
- **PCB fabrication completed**; hardware validation and testing are currently in progress.

RELEVANT COURSES

Electronics: Analog Electronics | Digital Systems | Circuit and Network Analysis | Semiconductor Fundamentals | PCB Design and Layout of Electronic Systems | Electrical Circuits Lab | Electronic Devices and Circuits Lab

Signal Processing: Digital Signal Processing | DSP Lab

Mathematics & Programming: Probability | Matrix Theory | Vector Calculus | Differential Equations | Introduction to Programming | Data Structures | Machine Learning

SKILLS

Programming Languages: Verilog | C | Python | MATLAB | LaTeX

Software & Tools: Xilinx Vivado | Quartus Prime Lite | Icarus Verilog | KiCad | Altium Designer | LTspice | Arduino IDE | GitHub

POSITIONS OF RESPONSIBILITY

Overall Head of Elektronica IITH Hyderabad

- Leading the Institute's electronics club, overseeing technical projects, competitions and collaborative events in digital design, embedded systems and signal processing related areas.
- Manage club operations and coordinate student teams and drive initiatives that promote engineering skills and innovation.

EXTRACURRICULAR

- **Active football player**, with a strong interest in team sports.
- **National Service Scheme (NSS) Volunteer**, actively participating in community service initiatives.